

Real and Fake Facial Recognition & XAI

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Overview

We analyzed two facial recognition deep neural networks (DNN) on a large number of images. Both mechanisms also used Grad-CAM as a separate addition for explainability.

Facial Detection mechanisms

- Use Resne18 and EfficientNet-B0 to analyze images.
- We used various datasets available on Kaggle to train the models.
- Total dataset volume was: (23.407345) GB.

XAI mechanism

- Which algorithm gives more coverage with Grad-CAM? It appears that EfficientNet-B0 does it better.

Models Example

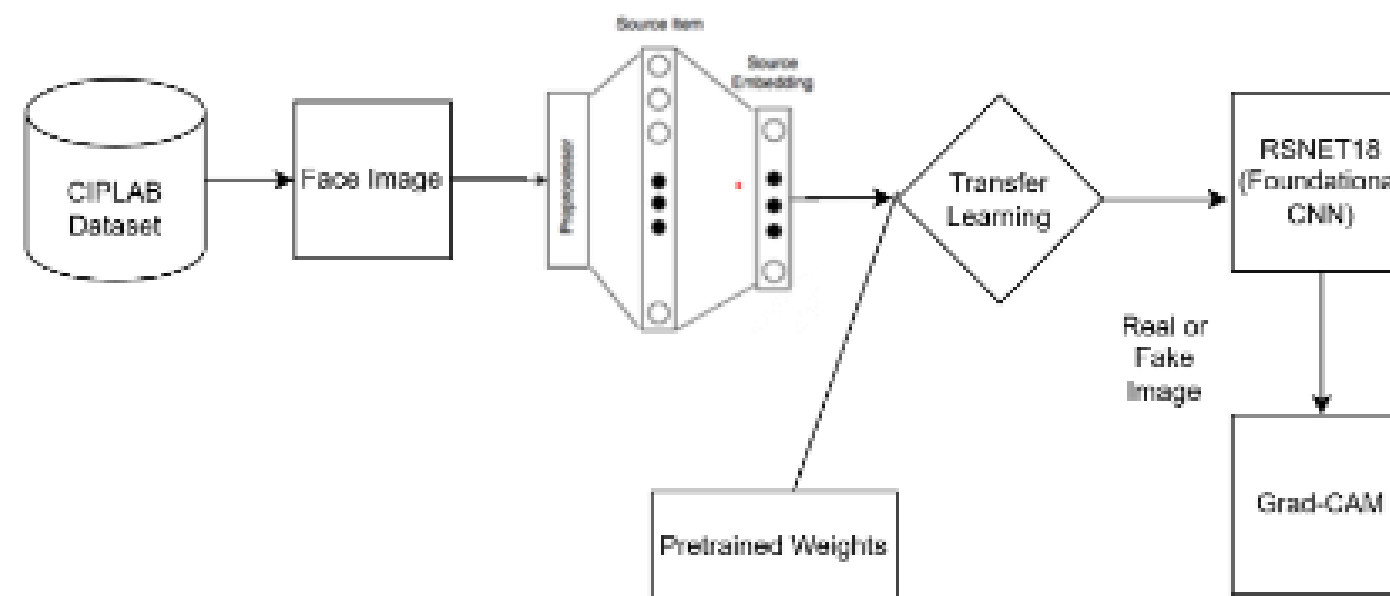


Fig. 1: Rsnets-18 training flow [1].

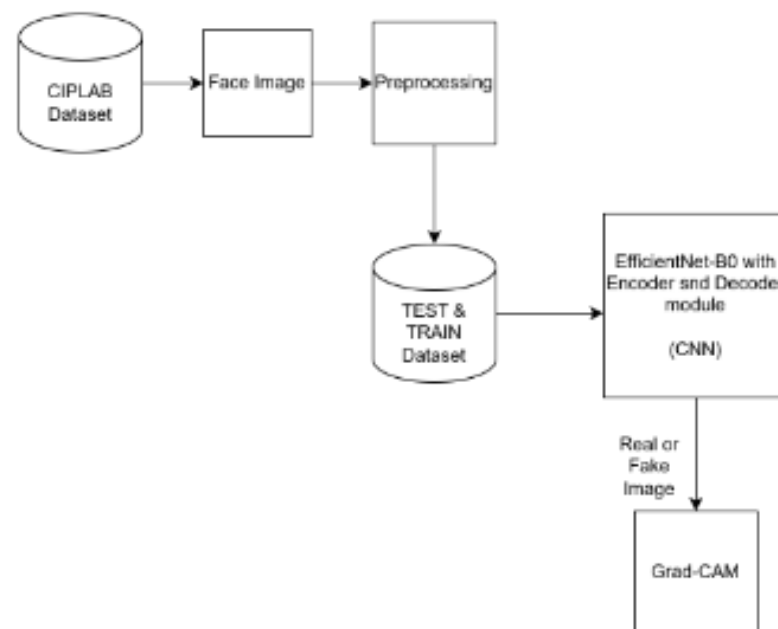


Fig. 2: EfficientNet-B0 training flow [2].

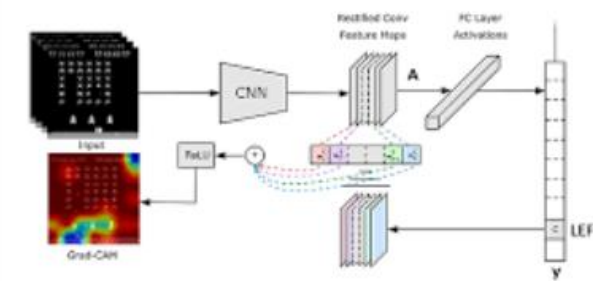


Fig. 3: Grad-CAM flow [3].

Discussion

- In EfficientNet-B0, it uses compound scaling to systematically balance a network's depth, width, and input resolution.

The Core Formula (Compound Scaling)

The entire EfficientNet family relies on scaling a baseline model using this single, uniform coefficient formula:

$$\phi = \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

Where:

- ϕ is a compound coefficient.
- α governs **depth** (number of layers).
- β governs **width** (number of channels).
- γ governs image **resolution** (input size).
- Subject to $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, meaning that for every twice (2x) increase in total computational resources, depth, width, and resolution scale harmoniously.

Fig. 4: EfficientNet-B0 Core Formula [4].

- ResNet-18 is a lightweight convolutional neural network that can be described using 3 core equations.

$$y = F(x, \{W_i\}) + x$$

$$y = F(x, \{W_i\}) + W_s x$$

$$p = \text{Softmax}(z)$$

Fig. 5: Rsnets-18 Core Formula [5].

Data

- Our facial recognition model is trained on the Real and Fake Face Detection from CIPLAB @ Yonsei University (Owner). License CC BY-NC-SA 4.0. It is available on Kaggle [6].
- We also used other Kaggle datasets. xhlulu/140k-real-and-fake-faces, mtesconi/twitter-deep-fake-text, manjilkarki/deepfake-and-real-images, sanikatiwarekar/deep-fake-detection-dfd-entire-original-dataset, saurabhbagchi/deepfake-image-detection, etc.



Fig 6: Resnet18 'fake' detection and grad-cam focus.



Fig 7: EfficientNet-B0 'fake' detection and grad-cam focus.

Model Accuracy

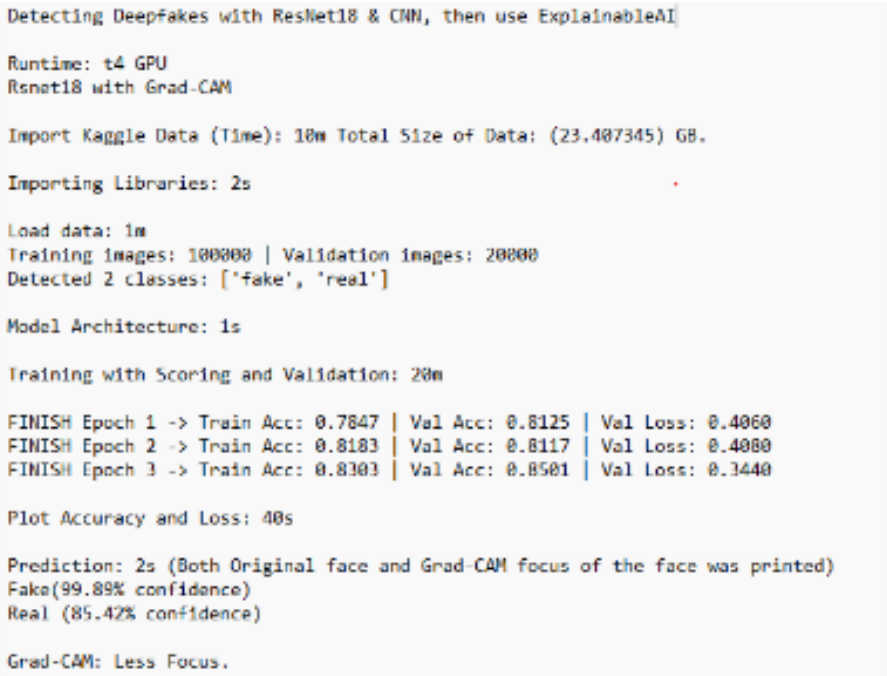


Fig 8: Rsnet-18 performance.

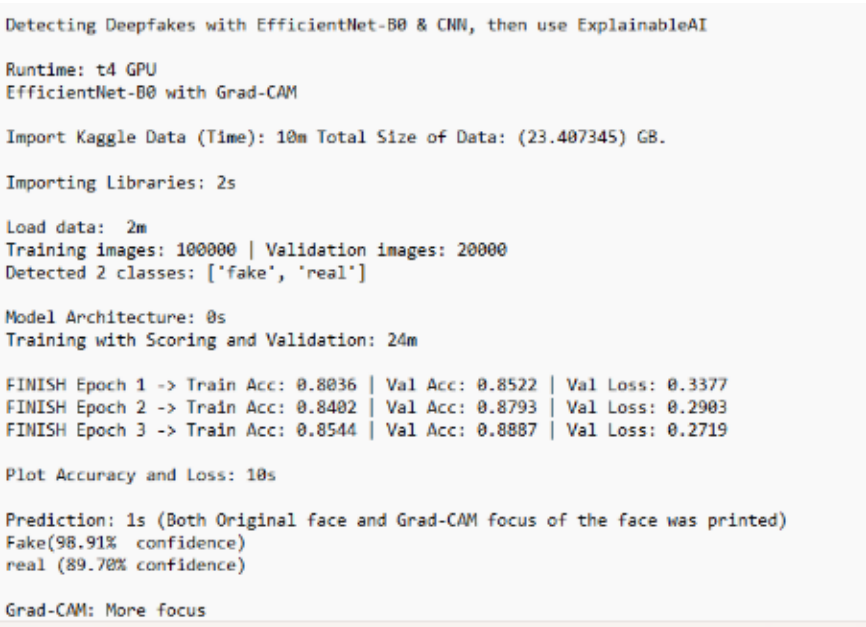


Fig 9: EfficientNet-B0 performance.

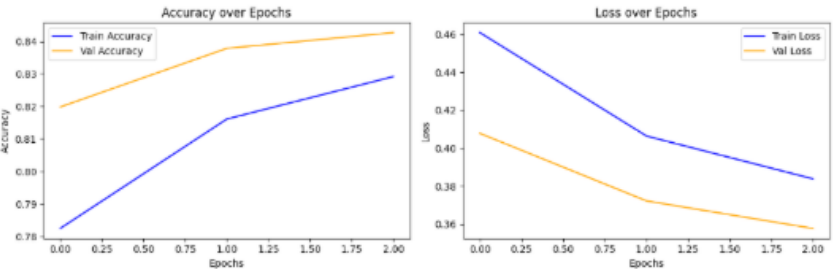


Fig 10: Rsnet-18 Accuracy and Loss.

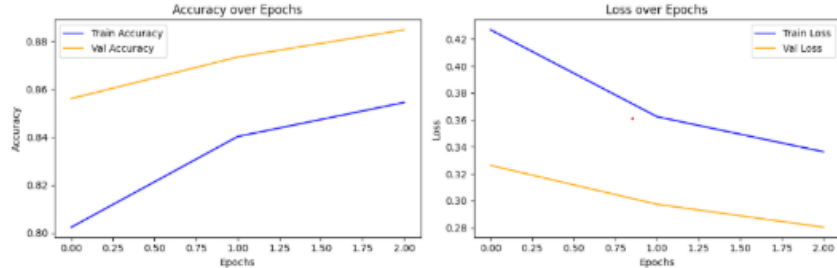


Fig 11: EfficientNet-B0 Accuracy and Loss.

Future Work

- Future work for ResNet18 and EfficientNet-B0 combined with Grad-CAM generally focuses on bridging the gap between high algorithmic accuracy and clinical or edge-deployment transparency. Because these lightweight convolutional neural networks (CNNs) are highly efficient, ongoing research continually refines how we interpret them [7].

References

1. Rsnet-18 diagram was modified by me. I used Stanford's Andrew Milich and Michael Karr's diagram as a baseline (Project name: Eluding Mass Surveillance: Adversarial Attacks on Facial Recognition Models).
2. EfficientNet architecture: https://www.researchgate.net/figure/Proposed-model-process-flow_fig2_357488896.
3. Grad-CAM architecture: <https://dl.acm.org/doi/pdf/10.1145/3723498.3725723>
4. www.google.com/url?sa=i&source=web&rct=j&url=https://medium.com/@kdk199604/efficientnet-smarter-not-just-bigger-neural-networks-94db3e2f8699&ved=2ahUKEwj6uc6lx9yUAXvxFjQIHdEuAl8Q0YISegolAggACAEIDhAC&opi=89978449&cd&psig=AOvVaw2uieXnl2Q0oXrPkhXAFUpQ&ust=1780077424213000
5. Sennan S, Somula R, Cho Y, Sennan S. A hybrid vision transformer and ResNet18 based model for biotic rice leaf disease detection. Front Plant Sci. 2025 Nov 14;16:1711700. doi: 10.3389/fpls.2025.1711700. PMID: 41323316; PMCID: PMC12661427.
6. Seonghyeon Nam, Seoung Wug Oh, Jae Yeon Kang, Chang Ha Shin, Younghyun Jo, Young Hwi Kim, Kyungmin Kim, Minho Shim, Sungho Lee, Yunji Kim, Suho Han, Gunhee Nam, Dasol Lee, Subin Jeon, In Cho, Woongoh Cho, Sejong Yang, Dongyoung Kim, Hyolim Kang, Sukjun Hwang, and Seon Joo Kim. (2019, January). Real and Fake Face Detection, Version 1. Retrieved [Date Retrieved] from <https://www.kaggle.com/datasets/ciplab/real-and-fake-face-detection>.
7. K, T., K, K. & Jeganathan, J. "A knowledge distillation framework integrating Grad-CAM in ResNet for imbalanced gastrointestinal abnormality classification in capsule endoscopy." *Discov Artif Intell* **6**, 359 (March 2026). <https://doi.org/10.1007/s44163-026-01081-x>.
8. Poster Template: UC Berkeley.